

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	593603
Award Program	2018 New Innovator in Food and Agriculture Research Award
Project Title	Developing CRISPR-Cpf1 genome editing technologies for crop improvement
Grant Start Date	10/01/2018
Grant End Date	03/31/2022
Total Grant Budget	\$565,685.55
Total FFAR Budget	\$282,842.78
Matching Funder(s)	\$282,842.78
Final Report Date	07/31/2022
Project Director/Principal Investigator Information	
Full Name	Yiping Qi
Email	yiping@umd.edu
Phone	301-405-7682
Grantee Organization Information	
Organization Name	University of Maryland, College Park
Address	4291 FIELDHOUSE DR
City, State Zip	College Park, MD, 20742
Tax ID	

Authorized Signing Official (ASO) Information	
Name	Stephanie Swartz
ASO Title	Contract Administrator
Email	sswartz1@umd.edu
Phone	301-405-8280

General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.

University of Maryland, College Park, MD 20742

- 1.2. How many new jobs were created because of this grant funding?

2

- 1.3. How many jobs were maintained because of this grant funding?

2

- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.

No.

- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.

No.

2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

Genome editing relies on the engineering of efficient and specific molecular scissors that can induce DNA double strand breaks in the genome which then initiate the editing process. CRISPR-Cpf1 is a

newly demonstrated DNA scissor that has enjoyed success in editing the genomes of plants, animals and humans. However, it has two major limitations which prevent its wide adoption in plant genome editing. First, its DNA targeting is limited to DNA sequences that contain three consecutive T (Thymine) nucleotides, which are not frequent in plant genomes. Second, no form of a Cpf1 nickase is available. A nickase only cuts one strand of the double-stranded DNA, which can promote many precise genome editing outcomes. Additionally, nickases can be further developed into base editing tools that will specifically introduce specific DNA base changes at desired genomic loci. A Cpf1 nickase system hence will be very useful in plant genome editing. This project leverages the funds from the New Innovator in Food and Agriculture Research Award at FFAR and matching funds from Syngenta to address these two major limitations of Cpf1. We will engineer Cpf1 variants that can target more genomic sites in a plant genome. We will also create novel Cpf1 nickase and base editing systems to enable precise genome editing. Our previous CRISPR tools have been distributed to over 200 labs around the world in the past three years. We will again make these new tools available to other scientists. It is anticipated that the outcome of this project will have enormous impact on food security, as the tools will significantly aid plant researchers and breeders to advance plant biology and crop improvement.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-wordlimit).

One goal of plant breeding is to introduce certain mutations and genes responsible for desirable traits into elite crop cultivars. In conventional breeding, the desired traits however, still must be introduced by genetic crossing, followed by selection among a large population of progeny. This process is tedious, expensive and time-consuming. For example, it can easily take over 10 years to develop an elite wheat variety with conventional breeding. When such a new variety is ready for release, it may become questionable whether it can function optimally in the changed climate. Clearly, the slow pace of conventional breeding will not meet the future demand of more food with increased human populations and decreased arable lands on our earth. With recent advances in science, it becomes very appealing to introduce desirable traits in elite crop backgrounds by directly editing the genome. In this way, plant breeding can be dramatically accelerated and crops with desired characteristics can be developed within a brief time. Plant genome editing represents a new breeding technology to increase our agricultural productivity and rise to meet challenges and opportunities in food security and climate change. In this project, New CRISPR-Cpf1 (currently Cas12a) tools will be developed including Cpf1 variants for broadened targeting scope and Cpf1 base editors for introducing pin-point mutations on the target genes of interest in crops. The technologies developed in this project will contribute to efficient and innovative breeding programs in many crops in the U.S. that are based on cutting-edge genome editing tools. It will help ensure food and nutrition security of U.S. Agriculture.

- 2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as "How did this project lead to improvements in U.S. agriculture and food systems?" or "How can the findings of this study guide or make improvement in future investigations and research?" Outcomes should be explained and classified in one of the following ways:

- a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;

Four major outcomes are generated from this project. They are four new group of genome editing tools for crop breeding, including 1) 6 novel Cpf1 (Cas12a) variants with high editing efficiency in plants such as MbCas12a with relaxed PAM requirements and with tolerance to low temperature; 2) a series of cytidine deaminases that can be used for efficient C-to-T base editing in crops with different editing windows; 3) efficient Cpf1 (Cas12a) based C-to-T base editors; 4) highly efficient Cpf1 (Cas12a) based A-to-G base editors.

- b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and

We have reported the novel CRISPR-Cpf1 (Cas12a) nucleases as new crop genome editing tools in multiple conferences and in a few peer-reviewed papers. The reagents have been disseminated to researchers and breeders in the U.S. through Addgene. We also published book chapters with detailed instructions to teach users on how to use these new genome editing reagents. Based on the feedback from many users, we learned the Cpf1 technologies resulting from our research have been rapidly adopted by many academic labs as well as a few biotech companies. For example, our work has provided key information and technology knowhow to major agro biotech companies in the U.S. including Bayer Crop Science, Inari Agriculture, and CTC Genomics.

- c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

Not relevant.

- 2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion

dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

There are two major goals of this project.

The first goal is to develop and engineer Cpf1 (Cas12a) proteins with relaxed PAM requirement for plant genome editing. The canonical PAM for Cpf1 is TTTV, which is restrictive for DNA targeting. We have successfully identified MbCas12a ortholog that can target relaxed VTTV PAMs, which hence broadens the targeting scope in a crop genome. This work has been included as a US patent application (US20210130838A1) filed by University of Maryland on 11/05/2020 with a priority date of 11/05/2019. The work has been wrapped up as Nature Communications paper published 03/29/2021. Through collaboration with IDT (Integrated DNA Technologies), we further engineered LbCas12a and developed LbCas12a-RVQ variant that improved genome editing at difficult sites including those sites with relaxed VTTV PAMs. This work has been finished and we are currently writing the manuscript for submission to Nature Communications.

The second goal is to develop Cpf1 (Cas12a) derived base editors in plants. Our original approach is to first engineer Cpf1 nickase and then engineer base editors using such a nickase. However, that route turned out to be challenging. To date, nobody has successfully engineered a Cpf1 nickase suitable for engineering base editors. So, we later decided to directly fuse cytidine deaminases and adenine deaminases to dCpf1 (de-activated Cpf1). We tried different Cpf1 variants, different deaminases, and different linkers that connect these two major components. Ultimately, we have successfully developed efficient Cpf1 based C-to-T base editors and very efficient Cpf1 A-to-G base editors that can readily introduce bi-allelic base changes on the target sites within the genes of interest in crops. Major experiments have been done by 03/30/2022. Right now, we are leverage other funding to finish the remaining experiments. To date, no peer-reviewed papers have reported efficient Cpf1 base editors. We hope to report these exciting reagents in a peer-reviewed journal soon and make the reagents available to others through Addgene.

- 2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

The two major goals have not been changed.

- 2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical

as the PI would like. (up to 5,000-word limit)

Toward Objective 1 on Screening and engineering Cpf1 proteins with relaxed PAM requirement for plant genome editing:

1) Major activities: We conducted protein engineering, molecular cloning of Cpf1 vectors and multiplexed expression systems, rice protoplast assay for transient expression of Cpf1 proteins and crRNAs for rapid assessment of genome editing efficiency, rice stable transformation for assessing genome editing outcomes and efficiency, high-order multiplex genome editing of rice plants, and assessment of dCpf1-SRDX based gene repression in stably transformed Arabidopsis plants.

2) Specific objectives: (1) screen Cpf1 variants for their utility in plant genome editing with different PAM requirements; (2) engineer Mb2Cpf1 (Mb2Cas12a) variants for improved genome editing efficiency at relaxed PAM sites; (3) develop highly multiplexable CRISPR-Cpf1 (Cas12a) systems for simultaneous editing of many trait genes in crops.

3) Significant results: (1) By screening 9 Cpf1 orthologs, we identified six novel Cpf1 proteins (Er, Lb5, Bs, Mb2, Ts, Mb) that confer efficient genome editing. Considering only two Cpf1 proteins (LbCpf1 and FnCpf1) were previously shown for reliable genome editing in plants, our work expanded the pool of useful Cpf1 proteins about 4 times larger. (2) We demonstrated that among all the novel Cpf1 proteins, Mb2Cpf1 can efficiently edit relaxed VTTV PAM sites. Furthermore, we engineered Mb2Cpf1-RVR variant and Mb2-Cpf1-RVRR variant to target TATV and more relaxed PAM sites. This greatly expanded targeting range of Cpf1/Cas12a in plant genomes. For example, our analysis showed that Mb2Cas12a-RVRR can edit 22% of the genome in rice, maize and wheat as compared to only 10% by the most popular CRISPR-Cas9 system which recognized the NGG PAM. (3) Mb2Cpf1 (Mb2Cas12a) can tolerate lower temperature during genome editing. This feature makes Mb2Cpf1 a promising genome editing reagent in agricultural crops since plant genome editing is done at relatively low temperature compared to human genome editing. (4) After comparing 10 multiplex Cas12a genome editing systems, we identified the best multiplex Cas12a/crRNA system where both Cas12a and crRNAs arrays are expressed by Pol II promoters and the crRNAs are processed by HH and HDV ribozymes. We demonstrated the system can simultaneously mutate 16 genes across 9 chromosomes in rice. Hence, this multiplex CRISPR/Cpf1 (Cas12a) system will allow for simultaneous breeding of many different agronomically important traits in a crop. Finally, we showed that such a multiplex Cpf1 system can be adapted for simultaneous silencing of multiple genes in plants using the CRISPR interference (CRISPRi) approach.

Toward Objective 2 on Engineering Cpf1 base editing systems for precise plant genome editing:

1) Major activities: We conducted protein engineering, molecular cloning, rice protoplast transformation, tomato protoplast transformation, and stable rice transformation to generate and assess Cpf1 base editing systems.

2) Specific objectives: (1) Compare different Cpf1 proteins for C-to-T base editing in rice; (2) Compare different Cpf1 proteins for A-to-G base editing in rice; (3) Compare different linkers with the best-performing Cpf1 to develop efficient C-to-T base editors; (4) Compare different linkers with the best-performing Cpf1 to develop efficient A-to-G base editors.

3) Significant results: (1) By comparing multiple Cpf1 proteins (FnCpf1, Mb2Cpf1, LbCpf1, ttLbCpf1), we developed ttLbCpf1 based C-to-T base editor and A-to-G base editor that can generate >50% editing efficiency in stably transformed rice plants. This makes Cpf1 base editors

as useful tools for precise genome editing in crops; (2) By comparing 10 different flexible linkers that connect the deaminases and Cpf1, we identified the best linkers for Cpf1 based C-to-T base editing using A3A/Y130F deaminase and the best linker for Cpf1 based A-to-G base editing using ABE8e deaminase in rice. (3) By comparing the top two best Cpf1 proteins, we identified LbCas12a-RVQ variant based base editors, when coupled with optimal linkers, generated highly efficient base editing, particularly with A-to-G base conversion at efficiency up to 100% in stable rice lines. With these data, we fulfilled our goals of this project, which is to develop efficient base editing tools for crops based on Cpf1.

- 2.7. Discuss efforts taken to ensure the approach is scientifically rigorous.
Include approaches taken to ensure robust and unbiased results.

To make sure our approaches are scientifically rigorous, we always include multiple target sites for testing editing efficiencies. For each target site, we will conduct three or more biological replicates in each experiment. For assessing editing frequency, we used next-generation sequencing, rather than Sanger sequencing, to precisely measure the efficiency in individual plants. We always include the necessary negative and positive controls, which is helpful for making appropriate conclusions. All our findings have been and will be published in reputable peer-reviewed journals, meaning they can sustain the scrutiny of top experts in the field.

- 2.8. List key stakeholders that could be served by results of this project.

The stakeholders for this project include scientists and breeders in both public and private sectors. The government agencies especially those in charge of biotechnologies and their regulation are also our stakeholders. Those in the general public who are interested in genome editing, biotechnology and crop improvement are also stakeholders.

- 2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research

sources will be reported under Information Products. (up to 1,000-word limit)

We have been using multiple means to disseminate our tools and knowledge resulting from this project. First, we deposit our CRISPR-Cpf1 base editing tools in the form of Golden Gate and Gateway cloning vectors to Addgene, a public vector repository. To date, Addgene has already disseminated ~2000 vectors from our lab to over 400 labs in more than 300 universities/institutes in 36 countries. Importantly, about half of the requests are from the scientists in the U.S. We have deposited some Cpf1 vectors resulting from this project and are on the way to deposit more that have not been published yet. These newest vectors will allow high-efficiency C-to-T and A-to-G base editing in crops for the first time. We expect these reagents will be heavily requested once they are made available up the publication of the studies. Second, we publish our tools with convincing demonstration in crops in the top-tier journals that involve rigorous peer review processes. When our tools are published in the top journals with high visibility, it helps the rapid recognition and adoption of these exciting technologies by others. Third, to teach many users who not versed in plant genome editing and/or molecular cloning or plant transformation, we nearly always will write a corresponding book chapter to describe the corresponding protocol in detail for each technology and have it published in the highly respected Methods in Molecular Biology Book series. For example, we have recently published such book chapters on multiplexed Cpf1 genome editing as well as on precise base editing in rice. Fourth, the PI (Dr. Qi) frequently present the work in conferences and invited talks in other institutions. Following such presentations, the PI often meet with the scientists who are interested in the technologies and answers their questions. Finally, the PI is very collaborative and willing to make the tools available to many others who wish to try them prior to the publishing of the tools. Such timely collaborations will enable beta testing of the technologies in other labs independently and importantly in many other plant species such as strawberry, maize, wheat, pear, apple, and tomato. The PI can gather useful feedback/data from these early users and make necessary improvement before releasing the more polished tools at Addgene to the public. This turned out to be an effective avenue to rapidly test and disseminate the technologies by leveraging the resources in collaborating labs in the research community.

- 2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

One technical challenge that we encountered is to engineer a Cpf1 nickase. Initially, we believed that we need to first engineer such a nickase before we can successfully develop any Cpf1 base editors with good editing efficiency. However, since Cpf1 only has one active nuclease domain that is responsible for the cleavage of both strands of DNA, it is very challenging to engineer a partly destroyed nuclease domain with specific cleavage activity on the targeting DNA strand, but not on the non-targeting strand. We tried some Cpf1 mutations but those always destroy Cpf1 nuclease activity completely. To date, we haven't seen any successful reports by others on a successful endeavor toward this, indicating the technical change to make such a protein. Later, when we engineered and identified Cpf1 variants (such as ttLbCpf1 and LbCpf1-RVQ) that have high DNA binding activity and developed highly active cytidine deaminase A3A/Y130F and highly

active adenine deaminase ABE8e, we realized that we probably do not need to rely on a Cpf1 nickase. Rather, using dCpf1 (de-activated Cpf1) may just be good enough. This alternative strategy indeed worked out nicely and we were able to develop Cpf1 C-to-T base editors and A-to-G base editors with high activity in plants.

- 2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

Two postdocs and one PhD students have been trained by working on the objectives in this project. They all have been trained with cutting edge genome editing tool development experiences. Their knowledge on Cpf1 and base editing plants has grown substantially, so that they are comfortable to serve peer reviewers to comment on manuscripts on these topics. They have coauthored multiple papers and book chapters in this area. One postdoc (Dr. Yingxiao Zhang) has exited our lab and she is now a Senior Scientist in Syngenta, a top agribiotech company, and she is leading a team of scientists to improve crops by using genome editing tools.

- 2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

This project has involved two postdoctoral scholars, one PhD student, one undergraduate student, and two high school interns.

3. Information Products

- 3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

The types of products include peer-reviewed publications and book chapters in Methods in Molecular Biology book series.

3.2. Please provide a list of citations for the information products produced during the project.

1. Yanhao Cheng[#], Simon Sretenovic[#], Yingxiao Zhang[#], Ji Huang, **Yiping Qi***. Expanding the targeting scope of FokI-dCas nuclease systems with SpRY and Mb2Cas12a. **Biotechnology Journal**. 2022 Apr 4;e2100571. <https://onlinelibrary.wiley.com/doi/10.1002/biot.202100571>
2. Shaun Curtin, **Yiping Qi**, Lázaro EP Peres, Alisdair R Fernie, Agustin Zsögön. Pathways to de novo domestication of crop wild relatives. **Plant Physiology**, 2021, 188 (4): 1746-1756. <https://doi.org/10.1093/plphys/kiab554>
3. Yingxiao Zhang[#], Yanhao Cheng[#], Hong Fang[#], Nathaniel Roberts, Liyang Zhang, Christopher A. Vakulskas, Randall P. Niedz, James N. Culver, **Yiping Qi***. Highly Efficient Genome Editing in Plant Protoplasts by Ribonucleoprotein Delivery of CRISPR-Cas12a Nucleases. **Frontiers in Genome Editing**, 2022. <https://doi.org/10.3389/fgeed.2022.780238>
4. Md. Mahmudul Hassan[†], Yingxiao Zhang^{#,†}, Guoliang Yuan, Kuntal De, Jin-Gui Chen, Wellington Muchero, Gerald A. Tuskan, **Yiping Qi***, Xiaohan Yang*. Construct design for CRISPR/Cas-based genome editing in plants. **Trends in Plant Science**, 2021. [https://www.cell.com/trends/plant-science/fulltext/S1360-1385\(21\)00157-6](https://www.cell.com/trends/plant-science/fulltext/S1360-1385(21)00157-6)
5. Kutubuddin Molla*, Simon Sretenovic[#], Kailash C Bansal, **Yiping Qi***. Precise plant genome editing with base editors and prime editors. **Nature Plants**, 2021, 7: 1166-1187. <https://www.nature.com/articles/s41477-021-00991-1>
6. Changtian Pan[#], Xincheng Wu^{##}, Kasey Markel, Aimee A. Malzahn[#], Neil Kundagrami^{##}, Simon Sretenovic[#], Yingxiao Zhang[#], Yanhao Cheng[#], Patrick M Shih, **Yiping Qi***. CRISPR-Act3.0 for highly efficient multiplexed gene activation in plants. **Nature Plants**, 2021. <https://www.nature.com/articles/s41477-021-00953-7>
7. Yingxiao Zhang^{#,†}, Qiurong Ren[†], Xu Tang[†], Shishi Liu, Aimee A. Malzahn[#], Jiaheng Wang, Desuo Yin[#], Changtian Pan[#], Mingzhu Yuan, Lan Huang, Han Yang, Yuxin Zhao, Qing Fang, Xuelian Zheng, Li Tian, Yanhao Cheng, Ysa Le^{##}, Bailey McCoy^{##}, Lidiya Franklin^{##}, Jeremy D. Selengut[#], Stephen M. Mount, Qiudeng Que, Yong Zhang*, **Yiping Qi***. Expanding the scope of plant genome engineering with novel Cas12a orthologs and highly multiplexable editing systems. **Nature Communications**, 2021. <https://www.nature.com/articles/s41467-021-22330-w>
8. Yingxiao Zhang[#], Brian Iaffaldano, **Yiping Qi***. CRISPR ribonucleoprotein-mediated genetic engineering in plants. **Plant Communications**, 2021, 2 (2): 100168. <https://doi.org/10.1016/j.xplc.2021.100168>
9. Changtian Pan[#], Simon Sretenovic[#], **Yiping Qi***. CRISPR/dCas-mediated transcriptional and epigenetic regulation in plants. **Current Opinion in Plant Biology**, 2021, 60, 101980. <https://doi.org/10.1016/j.pbi.2020.101980>

10. Kutubuddin A. Molla, **Yiping Qi**, Subhasis Karmakar, Mirza J. Baig. Base Editing Landscape Extends to Perform Transversion Mutation. *Trends in Genetics*, 2020, 36 (12): 899-901. <https://doi.org/10.1016/j.tig.2020.09.001>
11. Xu Tang, **Yiping Qi**, Yong Zhang. Single transcript unit CRISPR 2.0 systems for genome editing in rice. *Methods in Molecular Biology*. 2021, 2238: 193-204 https://link.springer.com/protocol/10.1007/978-1-0716-1068-8_12
12. Simon Sretenovic[#], Changtian Pan[#], Xu Tang, Yong Zhang, **Yiping Qi**^{*}. Rapid vector construction and assessment of BE3 and Target-AID C to T base editing systems in rice protoplasts. *Methods in Molecular Biology*. 2021, 2238: 95-113 https://link.springer.com/protocol/10.1007/978-1-0716-1068-8_7
13. Yingxiao Zhang[#], **Yiping Qi**^{*}. Diverse systems for efficient sequence insertion and replacement in precise plant genome editing. *BioDesign Research*, 2020. <https://doi.org/10.34133/2020/8659064>
14. Meiling Ming^{#,†}, Qiurong Ren[†], Changtian Pan^{#,†}, Yao He[†], Yingxiao Zhang^{#,†}, Shishi Liu, Zhaohui Zhong, Jiaheng Wang, Aimee A. Malzahn[#], Jun Wu, Xuelian Zheng, Yong Zhang^{*}, **Yiping Qi**^{*}. CRISPR-Cas12b enables efficient plant genome engineering. *Nature Plants*, 2020, 6(3):202-208. <https://www.nature.com/articles/s41477-020-0614-6>
15. Zhaohui Zhong[†], Shishi Liu[†], Xiaopei Liu, Binglin Liu, Xu Tang, Qiurong Ren, Jianping Zhou, Xuelian Zheng, **Yiping Qi**^{*}, Yong Zhang^{*}. Intron-based single transcript unit CRISPR systems for plant genome editing. *Rice*, 2020. <https://doi.org/10.1186/s12284-020-0369-8>
16. Filiz Gürel[#], Yingxiao Zhang[#], Simon Sretenovic[#], **Yiping Qi**^{*}. CRISPR-Cas nucleases and base editors for plant genome editing. *aBiotech*, 30 November 2019. <https://doi.org/10.1007/s42994-019-00010-0>
17. Shaoya Li, Yingxiao Zhang[#], Lanqin Xia^{*}, **Yiping Qi**^{*}. CRISPR-Cas12a enables efficient biallelic gene targeting in rice. *Plant Biotechnology Journal*. 2019. <https://doi.org/10.1111/pbi.13295>
18. Qiurong Ren, Zhaohui Zhong, Yan Wang, Qian Li, Mingzhu Yuan, Yao He, Qi Yu, Caiyan Qi, Xu Tang, Xuelian Zheng, Tao Zhang, **Yiping Qi**^{*}, Yong Zhang^{*}. Bidirectional promoter based CRISPR-Cas9 systems for plant genome editing. *Frontier in Plant Science*, 2019, <https://doi.org/10.3389/fpls.2019.01173>
19. Yingxiao Zhang[#], Aimee A Malzahn[#], Simon Sretenovic[#] and **Yiping Qi**^{*}. The emerging and uncultivated potential of CRISPR technology in plant science. *Nature Plants*, 2019, 5(8):778-794. <https://www.nature.com/articles/s41477-019-0461-5>
20. Yingxiao Zhang[#], **Yiping Qi**^{*}. CRISPR enables directed evolution in plants. *Genome Biology*. 2019 Apr 30;20(1):83. <https://genomebiology.biomedcentral.com/articles/10.1186/s13059-019-1693-4>
21. Aimee A. Malzahn^{†, #}, Xu Tang[†], Keunsub Lee, Qiurong Ren, Simon Sretenovic[#], Yingxiao Zhang[#], Hongqiao Chen, Minjeong Kang, Yu Bao, Xuelian Zheng, Kejun Deng, Tao Zhang, Valeria Salcedo[#], Kan Wang, Yong Zhang^{*}, **Yiping Qi**^{*}.

Application of CRISPR-Cas12a temperature sensitivity for improved genome editing in rice, maize, and Arabidopsis. **BMC Biology**. 2019, 17(1):9. <https://bmcbiol.biomedcentral.com/articles/10.1186/s12915-019-0629-5>

22. Xu Tang, Qiurong Ren, Lijia Yang, Yu Bao, Zhaohui Zhong, Binglin Liu, Yan Wang, Simon Sretenovic[#], Yingxiao Zhang[#], Xuelian Zheng, Tao zhang*, **Yiping Qi***, Yong Zhang*. Single transcript unit CRISPR 2.0 systems for robust Cas9 and Cas12a mediated plant genome editing. **Plant Biotechnology Journal**. 2019,17(7):1431-1445. <https://doi.org/10.1111/pbi.13068>

- 3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

Yes. See citations above.

- 3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

See publications above.

- 3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	1	0	US20210130838A1
Copyrights	0	0	
Trademarks	0	0	
Inventions	0	0	

- 3.6. Are any of the information products produced during this grant that are



confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

No.

- 3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that others can access and re-use these data in the future?

Please refer to my answers to "2.9". Briefly, in addition to using repositories, we have been and will continue to disseminate tools and information to the stakeholders through publishing peer-reviewed papers, book chapters, conference or institutional talks, and direct collaborations with interested parties.

4. Data Management

- 4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imagery). Please list the data generated for the award.

We published our results in peer-reviewed journals with associated data made available through those publications. There are not any other data (e.g., maps and imagery) generated.

- 4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

NA.

- 4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

NA.



FOUNDATION FOR
FOOD & AGRICULTURE
RESEARCH

Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: _____Yiping Qi_____

PI Signature: Yiping Qi _____ Date: 7/31/2022_____

Authorized Signing Official (ASO) Name: Stephanie Swartz, Sr. Contract Administrator, ORA

ASO Signature: Stephanie Swartz _____ Date: 09/29/22_____